

Homework 5 - Due to December 6th

Q1 -0

Q2 -10

Q3. good

Q4 -5

Exercise 1 (30 points). For each $a > 0$, find the equations of all the vertical and horizontal asymptotes, if they exist:

$$g(x) = \frac{\cos(x) \ln(x-a)}{\ln(e^x - e^a)}$$

First, we should find the domain of $g(x)$, that is

$$\begin{cases} x-a > 0 \\ e^x - e^a > 0 \end{cases} \Rightarrow x > a$$

Then, if we want to find vertical asymptotes, we should let $\ln(e^x - e^a) = 0$ that is $e^x - e^a = 1 \Rightarrow x = \ln(1 + e^a)$

When $x = \ln(1 + e^a)$, $\cos(\ln(1 + e^a)) \neq 0$, $\ln(\ln(1 + e^a) - a)$

Since $a \rightarrow 0$, $\ln(1 + e^a) \Rightarrow \ln e^a = a$, $\ln(1 + e^a) - a \Rightarrow a - a = 0$

$$\ln(x-a) \Rightarrow -\infty$$

In conclusion, $x = \ln(e^a + 1)$ is vertical asymptote.

To find horizontal asymptotes, we should consider $x \rightarrow \pm\infty$

When $x \rightarrow \infty$, $\ln(e^x - e^a) \Rightarrow \ln(e^x) = x$

$$\ln(x-a) \Rightarrow \ln x$$

$$\text{So } x \rightarrow \infty, g(x) = \frac{\cos(x) \ln x}{x}$$

Since $\cos(x) \in [-1, 1]$, $\frac{\ln x}{x} \rightarrow 0$, $g(x) \rightarrow 0$

In conclusion, when $x \rightarrow \infty$, $g(x) \rightarrow 0$

In conclusion, it's horizontal asymptote is $y = 0$

the vertical asymptote is $x = \ln(1 + e^a)$

Exercise 2 (30 points). Let $f(x) = \begin{cases} \sin^{2/3}(\pi x) & \text{if } -1 \leq x < 2 \\ 2 \tanh|x-3| & \text{if } x \geq 2 \end{cases}$

- (a) Find all the critical points, the intervals of increase and decrease and the local maximum and minimum values of f .
- (b) Find the points of the graph where f attains the absolute maximum and minimum values.

We can write derivative of $f(x)$ first,

$$f'(x) = \begin{cases} \frac{2}{3}\pi \sin^{-\frac{1}{3}}(\pi x) \cdot \cos(\pi x), & -1 \leq x < 2 \\ -2[1 - \tanh^2(3-x)] & , 2 \leq x < 3 \\ 2[1 - \tanh^2(x-3)] & , x \geq 3 \end{cases}$$

(a) ① $f(x) = \sin^{2/3}(\pi x)$ ($-1 \leq x < 2$)

Let $f'(x) = \frac{2}{3}\pi \sin^{-\frac{1}{3}}(\pi x) \cdot \cos(\pi x) = 0$

we can find $x = -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}$

When $-1 \leq x < -\frac{1}{2}$, $f'(x) > 0$, $f(x)$ increasing

$-\frac{1}{2} \leq x < 0$, $f'(x) \leq 0$, $f(x)$ decreasing

$0 \leq x < \frac{1}{2}$, $f'(x) > 0$, $f(x)$ increasing

$\frac{1}{2} \leq x < 1$, $f'(x) \leq 0$, $f(x)$ decreasing

$1 \leq x < \frac{3}{2}$, $f'(x) > 0$, $f(x)$ increasing

$\frac{3}{2} \leq x < 2$, $f'(x) \leq 0$, $f(x)$ decreasing

we can find $f(x)_{\min} = f(0) = \sqrt[3]{\sin^2 0} = 0$

$f(x)_{\max} = f(\frac{1}{2}) = \sqrt[3]{\sin^2 \frac{\pi}{2}} = 1$

② $f(x) = 2 \tanh|x-3|$ ($x \geq 2$) = $\begin{cases} 2 \tanh(3-x), & 2 \leq x < 3 \\ 2 \tanh(x-3), & x \geq 3 \end{cases}$

Let $f'(x) = 0$, that is $\begin{cases} -2[1 - \tanh^2(3-x)] = 0, & 2 \leq x < 3 \\ 2[1 - \tanh^2(x-3)] = 0, & x \geq 3 \end{cases}$

Since $(\tanh x)' = 1 - \tanh^2 x = \frac{4}{(e^x + e^{-x})^2} > 0$

That means we can't find x that $f'(x) = 0$

$-2[1 - \tanh^2(3-x)] < 0$, $2[1 - \tanh^2(x-3)] > 0$

When $2 \leq x < 3$, $f'(x) < 0$, $f(x)$ decreasing

$x \geq 3$, $f'(x) > 0$, $f(x)$ increasing

So $f(x)$ has no maximum, $f(x)_{\min} = f(3) = 0$

In conclusion, critical points are $x = -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}$

The increasing intervals are $[-1, -\frac{1}{2}]$, $[0, \frac{1}{2}]$, $[1, \frac{3}{2}]$, $[3, +\infty)$

The decreasing intervals are $(-\frac{1}{2}, 0)$, $(\frac{1}{2}, 1)$, $(\frac{3}{2}, 2)$, $(2, 3)$

The local maximum is $f(x) = 1$

local minimum is $f(x) = 0$

Exercise 2 (30 points). Let $f(x) = \begin{cases} \sin^{2/3}(\pi x) & \text{if } -1 \leq x < 2 \\ 2 \tanh|x-3| & \text{if } x \geq 2 \end{cases}$

- Find all the critical points, the intervals of increase and decrease and the local maximum and minimum values of f .
- Find the points of the graph where f attains the absolute maximum and minimum values.

(b) According to (a), we know the absolute minimum is $f(x) = 0$

$$\text{When } x \rightarrow \infty, \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} 2 \tanh(x-3) = 2 \lim_{x \rightarrow \infty} \frac{e^{x-3} - e^{-x}}{e^{x-3} + e^{-x}}$$

$$\text{Since } x \rightarrow \infty, e^{-x} \rightarrow 0, \text{ then we can write } 2 \lim_{x \rightarrow \infty} \frac{e^{x-3} - e^{-x}}{e^{x-3} + e^{-x}} \approx 2 \lim_{x \rightarrow \infty} \frac{e^{x-3}}{e^{x-3}} = 2$$

In conclusion, when $x \rightarrow \infty, f(x) \rightarrow 2$

Then, since when $2 \leq x < 3, f(x)$ decreasing, we should find $f(2)$

$$f(2) = 2 \tanh|2-3| = 2 \tanh 1 = 2 \cdot \frac{e^1 - e^{-1}}{e^1 + e^{-1}} = 2 \cdot \frac{e^2 - 1}{e^2 + 1} < 2$$

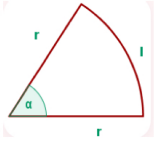
In conclusion, $f(x)$ has no absolute maximum

$x = -1, 0, 1, 3, f(x)$ has minimum

Exercise 3 (20 points). A circular sector has a perimeter of 10 cm.

Find the radius and the central angle corresponding to the sector with the largest area, enclosed by two radii and an arc.

Denote by l the length of the arc, by r the radius and by α the central angle (in radians).



We know the perimeter of circular sector is $2r + l = 2r + r\alpha = 10$ cm
the area of circular sector is $\frac{1}{2}lr = \frac{1}{2}\alpha r^2$

According to $r(2+\alpha) = 10$, $\alpha = \frac{10}{r} - 2$

$$\text{Then } \frac{1}{2}\alpha r^2 = \frac{1}{2}\left(\frac{10}{r} - 2\right)r^2$$

$$= \frac{1}{2}(10r - 2r^2)$$

$$= 5r - r^2 \quad (r > 0)$$

Assume $f(r) = 5r - r^2 \quad (r > 0)$

$$f'(r) = 5 - 2r$$

$$f'(r) = 0 \Rightarrow r = \frac{5}{2}$$

When $0 < r < \frac{5}{2}$, $f'(r) > 0$, $f(r)$ increasing

$r > \frac{5}{2}$, $f'(r) < 0$, $f(r)$ decreasing

In conclusion, $f(r)_{\max} = f\left(\frac{5}{2}\right)$

$$\text{When } r = \frac{5}{2}, \alpha = \frac{10}{\frac{5}{2}} - 2 = 2 \text{ rad}$$

In conclusion, when $r = \frac{5}{2}$ cm, $\alpha = 2$ rad, 
the circular sector has largest area.

Exercise 4 (20 points). Evaluate the definite integral $\int_0^3 (x^3 + 2x) dx$ using Riemann sums.

According to Riemann sums, we can divide $[0, 3]$ into n subintervals of equal width $\frac{3}{n}$, with endpoints given by

$$\{x_k = \frac{3k}{n}, 0 \leq k \leq n\}$$

we choose the left endpoints: $x_k^* = \frac{3k}{n}$

$$\int_0^3 (x^3 + 2x) dx = \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{k=0}^n \left[\left(\frac{3k}{n} \right)^3 + 2 \cdot \frac{3k}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{k=0}^n \left[\frac{27k^3}{n^3} + \frac{6k}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \sum_{k=0}^n k^3 + \frac{18}{n^2} \cdot \sum_{k=0}^n k \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \frac{1}{4} n^2 (n+1)^2 + \frac{18}{n^2} \cdot \frac{n(n+1)}{2} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \cdot \frac{n^4}{4} \left(1 + \frac{1}{n}\right)^2 + 9 \cdot \left(1 + \frac{1}{n}\right) \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{81}{4} \left(1 + \frac{1}{n}\right)^2 + 9 \left(1 + \frac{1}{n}\right) \right]$$

$$= \frac{81}{4} + 9$$

$$= \frac{107}{4}$$

In conclusion, $\int_0^3 (x^3 + 2x) dx = \frac{107}{4}$

Index of comments

2.1 Many critical points are missing